Experiment Number : 04 **Date: 05/02/2026**

ARRAYS

Aim:

To understand and implement array operations in Java.

PRE LAB EXERCISE

QUESTIONS

- ✓ What is an array?

  An array is a collection of elements of the same data type stored in contiguous memory locations. Each element can be accessed using an index value.

- ✓ Why are arrays used?

  Arrays are used to:

  Store multiple values of the same type using a single variable

  Reduce code complexity

  Allow fast access to elements using index

  Improve memory organization and efficiency

- ✓ What is the difference between array and variable?

Variable	Array
Stores only one value	Stores multiple values
Uses a single memory location	Uses multiple contiguous memory locations
Accessed by variable name	Accessed using index
Simple data storage	Used for bulk data handling

IN LAB EXERCISE

Objective:

To perform array operations using simple programs.

PROGRAMS:

1. Program to Read and Print Array Elements

Code:

```
import java.util.Scanner;
public class ReadPrintArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] arr = new int[5];
        System.out.println("Enter 5 elements:");
        for(int i = 0; i < 5; i++)
            arr[i] = sc.nextInt();
        System.out.println("Array elements are:");
        for(int i = 0; i < 5; i++)
            System.out.print(arr[i] + " ");
    }
}
```

OUTPUT:

Input:
10 20 30 40 50

Output:

Array elements are:

10 20 30 40 50

module-info.java | ReadPrintArray.java

```java
package ex_4;

import java.util.Scanner;
public class ReadPrintArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] arr = new int[5];
        System.out.println("Enter 5 elements:");
        for(int i = 0; i < 5; i++)
            arr[i] = sc.nextInt();
        System.out.println("Array elements are:");
        for(int i = 0; i < 5; i++)
            System.out.print(arr[i] + " ");
    }
}
```

Problems | Javadoc | Declaration | Console | Eclipse IDE for Java Dev

<terminated> ReadPrintArray [Java Application] C:\Users\Sarum\Downloads\eclipse-java-2

```
Enter 5 elements:
1
2
3
4
5
Array elements are:
1 2 3 4 5
```

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2. Program to Find Sum of Array Elements

Code:

```
import java.util.Scanner;
public class SumArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] arr = new int[5];
        int sum = 0;
        System.out.println("Enter 5 elements:");
        for(int i = 0; i < 5; i++)
            arr[i] = sc.nextInt();
        for(int i = 0; i < 5; i++)
            sum += arr[i];
        System.out.println("Sum = " + sum);
    }
}
```

OUTPUT:

Input:
5 10 15 20 25

Output:

Sum = 75

module-info.java SumArray.java

```java
package ex_4;

import java.util.Scanner;
public class SumArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] arr = new int[5];
        int sum = 0;
        System.out.println("Enter 5 elements:");
        for(int i = 0; i < 5; i++)
            arr[i] = sc.nextInt();
        for(int i = 0; i < 5; i++)
            sum += arr[i];
        System.out.println("Sum = " + sum);
    }
}
```

Problems @ Javadoc Declaration Console Eclipse IDE for

<terminated> SumArray [Java Application] C:\Users\Sarum\Downloads\eclipse-ja

```
Enter 5 elements:
5
10
15
20
25
Sum = 75
```

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3. Program to Find Largest Element in an Array

Code:

```
import java.util.Scanner;
public class LargestElement {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] arr = new int[5];
        System.out.println("Enter 5 elements:");
        for(int i = 0; i < 5; i++)
            arr[i] = sc.nextInt();
        int max = arr[0];
        for(int i = 1; i < 5; i++)
            if(arr[i] > max)
                max = arr[i];
        System.out.println("Largest element = " + max);
    }
}
```

OUTPUT:

Input:

12 45 23 9 30

Output:

Largest element = 45

module-info.java LargestElement.java

```java
package ex_4;

import java.util.Scanner;
public class LargestElement {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] arr = new int[5];
        System.out.println("Enter 5 elements:");
        for(int i = 0; i < 5; i++)
            arr[i] = sc.nextInt();
        int max = arr[0];
        for(int i = 1; i < 5; i++)
            if(arr[i] > max)
                max = arr[i];
        System.out.println("Largest element = " + max);
    }
}
```

Problems @ Javadoc Declaration Console Eclipse IDE for Jav

<terminated> LargestElement [Java Application] C:\Users\Sarum\Downloads\eclipse-

```
Enter 5 elements:
12
45
23
9
30
Largest element = 45
```

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4. Program to Reverse an Array

Code:

```
import java.util.Scanner;


public class ReverseArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] arr = new int[5];
        System.out.println("Enter 5 elements:");
        for(int i = 0; i < 5; i++)
            arr[i] = sc.nextInt();
        System.out.println("Reversed array:");
        for(int i = 4; i >= 0; i--)
            System.out.print(arr[i] + " ");
    }
}
```

OUTPUT:

Input:
1 2 3 4 5

Output:

Reversed array:

5 4 3 2 1

module-info.java ReverseArray.java

```java
package ex_4;

import java.util.Scanner;

public class ReverseArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] arr = new int[5];
        System.out.println("Enter 5 elements:");
        for(int i = 0; i < 5; i++)
            arr[i] = sc.nextInt();
        System.out.println("Reversed array:");
        for(int i = 4; i >= 0; i--)
            System.out.print(arr[i] + " ");
    }
}
```

Problems @ Javadoc Declaration Console Eclipse ID

<terminated> ReverseArray [Java Application] C:\Users\Sarum\Downloads\e

```
Enter 5 elements:
1
2
3
4
5
Reversed array:
5 4 3 2 1
```

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5. Program to Count Even and Odd Numbers

Code:

```
import java.util.Scanner;
public class EvenOddCount {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] arr = new int[5];
        int even = 0, odd = 0;
        System.out.println("Enter 5 elements:");
        for(int i = 0; i < 5; i++)
            arr[i] = sc.nextInt();
        for(int i = 0; i < 5; i++) {
            if(arr[i] % 2 == 0)
                even++;
            else
                odd++;
        }

        System.out.println("Even = " + even);
        System.out.println("Odd = " + odd);
    }
}
```

OUTPUT:

Input:
2 7 4 9 10

Output:

Even = 3

Odd = 2

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module-info.java EvenOddCount.java

```java
package ex_4;

import java.util.Scanner;
public class EvenOddCount {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] arr = new int[5];
        int even = 0, odd = 0;
        System.out.println("Enter 5 elements:");
        for(int i = 0; i < 5; i++)
            arr[i] = sc.nextInt();
        for(int i = 0; i < 5; i++) {
            if(arr[i] % 2 == 0)
                even++;
            else
                odd++;
        }
```

Problems @ Javadoc Declaration Console Eclipse IDE for Java

<terminated> EvenOddCount [Java Application] C:\Users\Sarum\Downloads\eclipse-ja

```
Enter 5 elements:
1
2
3
4
5
Even = 2
Odd = 3
```

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6. Program to Sort Array in Ascending Order

Code:

```
import java.util.Scanner;
public class SortArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] arr = new int[5];
        int temp;
        System.out.println("Enter 5 elements:");
        for(int i = 0; i < 5; i++)
            arr[i] = sc.nextInt();
        for(int i = 0; i < 5; i++) {
            for(int j = i + 1; j < 5; j++) {
                if(arr[i] > arr[j]) {
                    temp = arr[i];
                    arr[i] = arr[j];
                    arr[j] = temp;
                }
            }
        }
        System.out.println("Sorted array:");
        for(int i = 0; i < 5; i++)
            System.out.print(arr[i] + " ");
    }
}
```

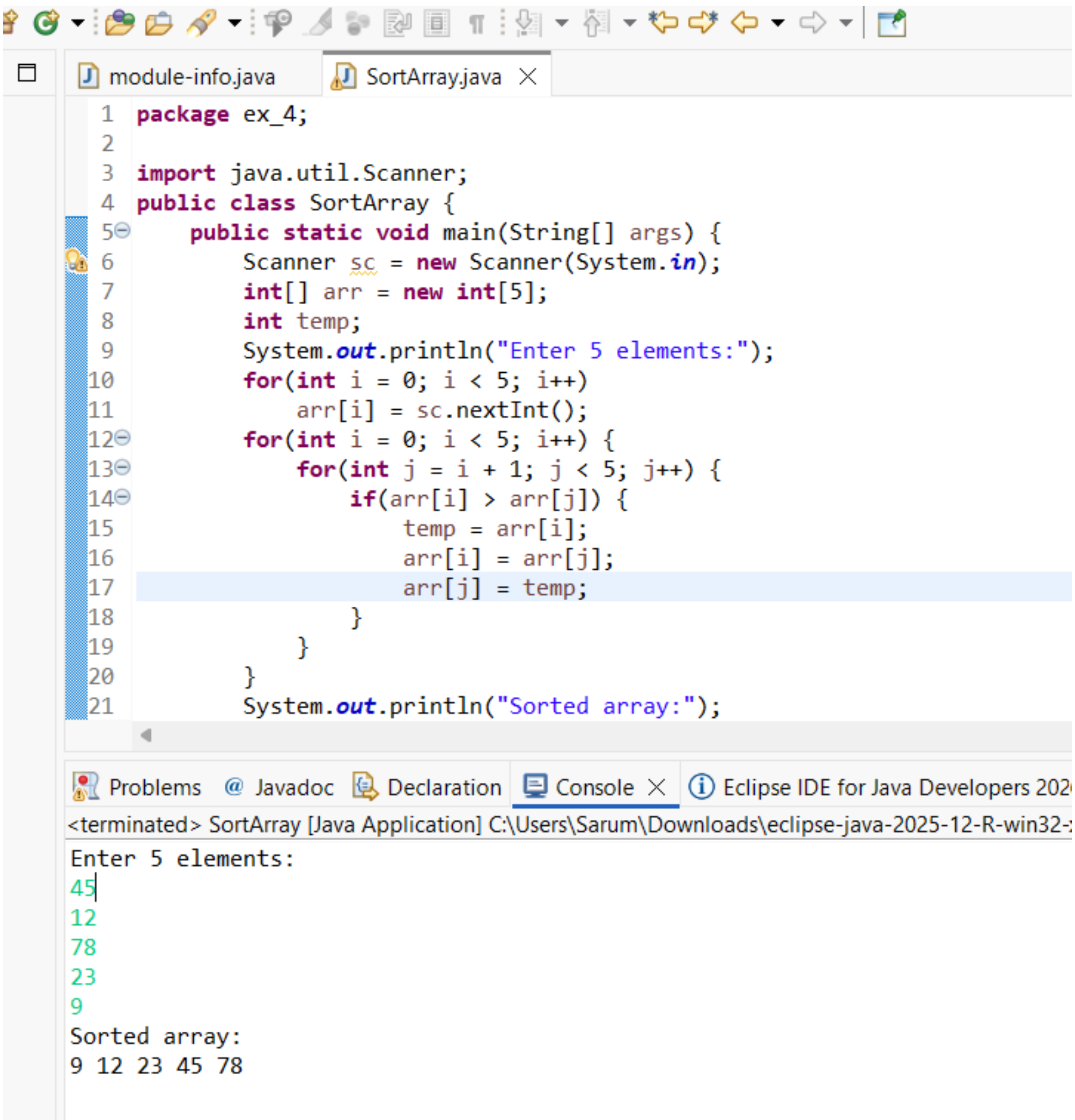
OUTPUT:

Input:
45 12 78 23 9

Output:

Sorted array:

9 12 23 45 78



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7. Program to Find Second Largest Element

Code:

```
import java.util.Scanner;
public class SecondLargest {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] arr = new int[5];

        System.out.println("Enter 5 elements:");
        for(int i = 0; i < 5; i++)
            arr[i] = sc.nextInt();
        int largest = arr[0];
        int second = arr[0];
        for(int i = 0; i < 5; i++) {
            if(arr[i] > largest) {
                second = largest;
                largest = arr[i];
            }
        }
        System.out.println("Second largest = " + second);
    }
}
```

OUTPUT:

Input:
10 45 23 89 67

Output:

Second largest = 67

module-info.java   SecondLargest.java

```java
package ex_4;

import java.util.Scanner;

public class SecondLargest {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] arr = new int[5];

        System.out.println("Enter 5 elements:");
        for(int i = 0; i < 5; i++)
            arr[i] = sc.nextInt();

        int largest = arr[0];
        int second = arr[0];

        for(int i = 1; i < 5; i++) {
            if(arr[i] > largest) {
                second = largest;
                largest = arr[i];
            }
```

Problems   Javadoc   Declaration   Console   Eclipse IDE fc

<terminated> SecondLargest [Java Application] C:\Users\Sarum\Downloads\ecli

```
Enter 5 elements:
10
45
23
89
67
Second largest = 67
```

8. Program for Matrix Addition (2D Array)

Code:

```
import java.util.Scanner;
public class MatrixAddition {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[][] a = new int[2][2];
        int[][] b = new int[2][2];
        int[][] sum = new int[2][2];
        System.out.println("Enter elements of matrix A:");
        for(int i = 0; i < 2; i++)
            for(int j = 0; j < 2; j++)
                a[i][j] = sc.nextInt();
        System.out.println("Enter elements of matrix B:");
        for(int i = 0; i < 2; i++)
            for(int j = 0; j < 2; j++)
                b[i][j] = sc.nextInt();
        for(int i = 0; i < 2; i++)
            for(int j = 0; j < 2; j++)
                sum[i][j] = a[i][j] + b[i][j];
        System.out.println("Sum matrix:");
        for(int i = 0; i < 2; i++) {
            for(int j = 0; j < 2; j++)
                System.out.print(sum[i][j] + " ");
            System.out.println();
```

```
        }
    }
}
```

OUTPUT:

Matrix A:

1 2

3 4

Matrix B:

5 6

7 8

Sum matrix:

6 8

10 12

module-info.java   MatrixAddition.java

```java
package ex_4;


import java.util.Scanner;
public class MatrixAddition {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[][] a = new int[2][2];
        int[][] b = new int[2][2];
        int[][] sum = new int[2][2];
        System.out.println("Enter elements of matrix A:");
        for(int i = 0; i < 2; i++)
            for(int j = 0; j < 2; j++)
                a[i][j] = sc.nextInt();
        System.out.println("Enter elements of matrix B:");
        for(int i = 0; i < 2; i++)
            for(int j = 0; j < 2; j++)
                b[i][j] = sc.nextInt();
        for(int i = 0; i < 2; i++)
            for(int j = 0; j < 2; j++)
                sum[i][j] = a[i][j] + b[i][j];
```

Problems   @ Javadoc   Declaration   Console   Eclipse IDE for Java Develo

<terminated> MatrixAddition [Java Application] C:\Users\Sarum\Downloads\eclipse-java-202

```
Enter elements of matrix A:
1 2
 3 4
Enter elements of matrix B:
5 6
7 8
Sum matrix:
6 8
10 12
```

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POST LAB EXERCISE

- ✓ Why is array indexing usually started from zero instead of one?

  Array indexing starts from zero because the index represents the offset from the base memory address. The first element is stored at the base address itself, so its offset is zero.

- ✓ What happens if we try to access an array element outside its declared size?

  If we access an array element outside its size, Java throws an ArrayIndexOutOfBoundsException at runtime.

- ✓ How does memory allocation differ for static arrays and dynamic arrays?

  Static arrays: Size is fixed at compile time and cannot be changed.
  Dynamic arrays: Size is decided at runtime and can be changed (e.g., using ArrayList).

- ✓ Why is searching faster in arrays compared to linked lists?

  Arrays support direct access using index, so elements can be accessed in O(1) time.
  Linked lists require sequential traversal, which takes more time.

- ✓ What is the difference between contiguous and non-contiguous memory allocation?

Contiguous Memory	**Non-Contiguous Memory**
Memory blocks are adjacent	Memory blocks are scattered
Used in arrays	Used in linked lists
Faster access	Slower access
Fixed size	Dynamic size

Result:

Thus the array operations were executed successfully.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	**5**	
In Lab Exercise	**10**	
Post Lab Exercise	**5**	
Viva	**10**	
Total	**30**	
Faculty Signature		